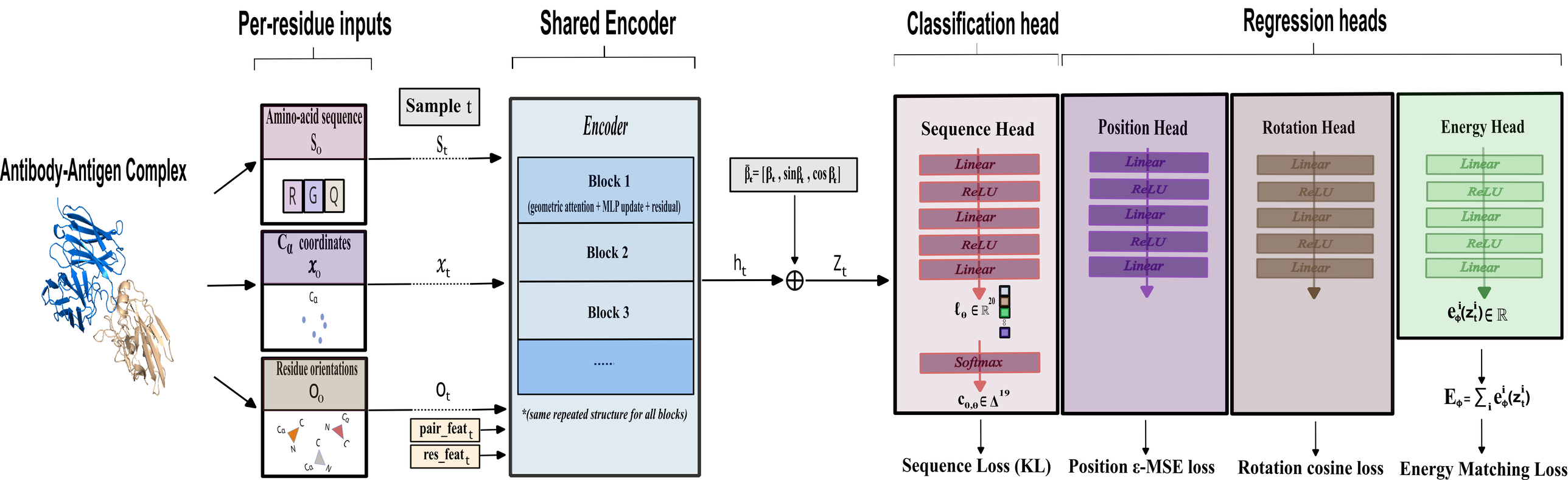
Per-residue inputs
Shared Encoder
Classification head
Regression heads
Antibody-Antigen Complex
Amino-acid sequence
S_0
R G Q
C_α coordinates
x_0
Residue orientations
O_0
Sample t
S_t
x_t
O_t
pair_feat$_t$
res_feat$_t$
Encoder
Block 1
(geometric attention + MLP update + residual)
Block 2
Block 3
.....
*(same repeated structure for all blocks)
$\bar{\beta}_t = [\beta_t, \sin\beta_t, \cos\beta_t]$
h_t
Z_t
Sequence Head
Linear
ReLU
Linear
ReLU
Linear
$\ell_\theta \in \mathbb{R}^{20}$
Softmax
$c_{0,\theta} \in \Delta^{19}$
Position Head
Linear
ReLU
Linear
ReLU
Linear
Rotation Head
Linear
ReLU
Linear
ReLU
Linear
Energy Head
Linear
ReLU
Linear
ReLU
Linear
$e^i_\phi(z^i_t) \in \mathbb{R}$
$E_\phi = \sum_i e^i_\phi(z^i_t)$
Sequence Loss (KL)
Position ε-MSE loss
Rotation cosine loss
Energy Matching Loss